



Transfer matrices to characterize linear and quadratic acoustic black holes in duct terminations



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ABSTRACT

The acoustic black hole (ABH) effect for sound reduction in duct terminations can be accomplished by means of retarding structures. The latter act as waveguides and their performance relies on two factors. First, a power-law decay of the duct radius and, second, an appropriate dependence of the wall admittance with the duct radius. In theory, the ABH can be achieved placing a set of rigid rings inside the duct, with inner radii and inter-spacing decreasing to zero as the tube end section is approached. In this work we focus on the linear and quadratic inner radius decay cases, referred to as the linear and quadratic ABHs. To begin with, analytical expressions are derived for the quadratic ABH and compared to those of the linear one. In both cases the solutions become singular at the final section of the duct. The wall admittance manifests the same behavior. Therefore, one has to deal with imperfect ABHs ending before the singularity, even in the best case scenario. Yet in practice, one may encounter further factors that deteriorate the ABH behavior. The number of rings and cavities between them is finite as it is the ring thicknesses. Damping also plays an important role. It is herein proposed to analyze the influence of all these factors on the reflection coefficient of the ABHs by means of the transfer matrix method (TMM). Transfer matrices are presented which allow one to relate the acoustic pressure and acoustic particle velocity between different sections of the retarding structure. They constitute a quick and valuable tool for an initial design of ABHs.

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1. Introduction

The acoustic black hole (ABH) effect is a passive approach for the control of vibrations and sound (see e.g., the recent review in [1]). The effect is achieved by means of a retarding structure (or a tailored geometry). Such a structure induces a power-law-like decay of the velocity of the waves that propagate in it. In theory, an incident wave will become trapped and never reach the edge of the structure. The propagation wave velocity tends to zero with distance in such a way that a wave would spend an infinite amount of time to get to the boundary. Therefore, no reflection can occur from there which motivates the term “acoustic black hole”.

To date, most efforts have been placed on ABHs for flexural waves in beams and plates. One-dimensional ABHs can be achieved by varying the beam and/or plate edge thickness following a power law profile. The idea was first proposed in [2] and became more popular with the works in [3–5]. In practice, a perfect ABH effect cannot be manufactured because it

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would require a wedge of infinite extent. However, as quoted in [3,4,6], one can partially compensate for the imperfection of a finite thickness edge by covering the tip of the profile with a damping layer. Yet this is not the sole type of imperfection one may encounter in real implementations of ABHs. For instance, the effects of deformations due to wedge attachments to plates [7], or due to stress relaxation when manufacturing the ABHs [8], are of practical importance and have been recently analyzed in the referenced works. The ABHs have been proved to be quite robust to these imperfections, still providing low reflection coefficients. On the other hand, it is worthwhile mentioning that two-dimensional ABHs have also been designed. These usually consist of plates with tapered holes having an appropriate inner profile [9–11].

As opposed to ABHs for beams and plates, less attention has been paid to the possibility of using the acoustic black hole effect for sound reduction. The first proposal in that direction was presented in [12]. A theoretical analysis was made for a retarding structure consisting of a set of rings and cavities placed at the termination of a duct. The approximated wall admittance of that structure was such that when combined with a power-law decaying radius for the inner rings resulted in an ABH effect, with no sound waves reflecting from the end of the duct. Practical realizations of the waveguide in [12] for linear and quadratic decaying ring radii have been recently attempted in [13,14]. Besides, other possibilities for one-dimensional sound reducing ABHs are suggested in [1]. Two-dimensional designs of ABH sound absorbers have been also lately proposed in analogy to optical black holes. These usually consist of a graded index metamaterial shell that acts as an impedance matching layer between air and an inner porous absorber core [15,16], or second metamaterial [17], which dissipates acoustic energy. This results in an omnidirectional broadband absorbing device.

This paper focuses on the ABH for ducts suggested in [12], which can be seen as a particular case of a reactive anechoic termination. Anechoic terminations are of practical importance in the design of wind tunnels, fan and propeller test rigs, and for the acoustic characterization of mufflers. The measurement of mufflers' transmission loss, for instance, usually relies on the so-called two-microphone transfer function method [18], or some of its enhanced variations [19,20]. The corresponding setup requires a duct with an anechoic termination. Research on them have been going on for decades and very different approaches have been attempted, which comprise from using wedges [21–23], absorbing layers [24,25], conical, exponential and catenoidal profiles for smoothing the impedance mismatch [26,27], using micro-perforated panels [28,29] or resorting to active noise control strategies [30]. The ABH may be viewed as an alternative strategy to achieve anechoicity.

In this work we are interested in the particular cases of linear and quadratic decaying ring radii for the retarding structures in [12], which will be hereafter referred to as the linear ABH and the quadratic ABH, following the nomenclature in [13,14]. To begin with, we will develop analytical expressions for the quadratic ABH to complement those for the linear one in [12]. It turns out that the acoustic pressure of the linear and quadratic ABHs becomes singular at the duct end section. The same happens to the wall admittance. As a consequence, even for theoretical computations of the waveguide reflection coefficient, one has to assume an imperfect ABH that finishes before the end section of the duct. Yet, when building an ABH (see [13,14]) that is not the sole source of limitations. It is also necessary to deal with a finite number of cavities, separated by rings of finite thickness, which follow a certain spatial distribution. The admittance of such a realistic waveguide departs from the theoretical one and thus restrains the ABH effect. The situation is somewhat akin to that encountered in structural ABHs, where the potential decaying wedges should have infinite extent but have to be truncated for practical implementations.

A precise analysis of the linear and quadratic ABHs could be carried out by means of finite element simulations (FEM). However, FEM can be time consuming and costly if one has to check a large number of configurations for varying parameters. Therefore, prior to FEM it is herein proposed to make use of the transfer matrix method (TMM, see e.g., [31]) for acoustic filters. The ABH retarding structure is approximated as a duct of varying cross-section with several branch cavities appended to it. The relation between state vectors of acoustic pressure and acoustic particle velocity at different sections of the waveguide is then established by means of analytical transfer matrices. The TMM enables one to quickly analyze the parameters influencing the performance of the retarding structure, such as the number and size of rings and cavities, their location, the damping, the ring thicknesses, etc. Once the dependence of the ABHs on those parameters is made clear, one could chose a small set of the most promising configurations for a more detailed FEM analysis (FEM simulations are, however, out of the scope of this paper and will be carried out in future work). The TMM has been traditionally applied in many areas of acoustics. To mention a few, these comprise from the design of mufflers [31], to that of musical instruments [32,33] or to articulatory speech synthesis [34].

The paper is organized as follows. Section 2 first presents the governing equation of the ABH. Then, the theory behind the linear ABH in [12] is reviewed for completeness (though presented in a slightly different way) and analogous expressions get derived for the quadratic ABH. In Section 3, transfer matrices are constructed and combined to deal with realistic exemplifications of the linear and quadratic ABH retarding structures. Comparisons between the theoretical and realistic ABHs are presented in Section 4, together with the dependence on several design parameters. The conclusions close the paper in Section 5.

2. The acoustic black hole effect in duct terminations

2.1. Governing equation

The equation that governs plane wave propagation in an axisymmetric waveguide of varying cross section $S(x)$ and wall admittance $Y(x)$ can be determined from the linearized continuity and momentum conservation equations. The result is a

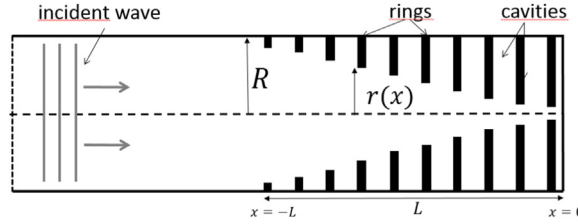


Fig. 1. Sketch of the retarding structure that produces the ABH effect.

slight generalization of Webster's equation. Assuming a time harmonic dependence of the acoustic pressure, $p(x, t) = p(x)\exp(-j\omega t)$, the equation reads in the frequency domain (see [12]),

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x} \frac{\partial(\ln S)}{\partial x} + p \left[k_0^2 + jZ_0 k_0 \frac{2Y}{r} \right] = 0, \quad (1)$$

where x denotes the axial coordinate with origin $x=0$ at the duct termination and $r(x)$ stands for the waveguide radius. As usual, $k_0 = \omega/c_0$ represents the wavenumber, ω being the radial frequency and c_0 the speed of sound. $Z_0 = \rho_0 c_0$ denotes the characteristic impedance of air and ρ_0 standing for its density.

We address the case of a rigid circular duct of constant radius R and place a retarding structure at its termination end, starting at $x = -L$ and ending at $x=0$ (see Fig. 1). The goal of such a structure is that waves impinging on it experience no reflection at all. This ABH effect can be achieved if the propagation speed of a wave decreases as it approaches the end of the duct, in such a way that it would take an infinite amount of time for it to reach $x=0$. To that purpose we consider the retarding structure in [12] consisting of a set of rigid rings with decreasing radius $r(x)$, mounted inside the duct and separated by air cavities. Two requisites can be shown to be necessary to achieve the ABH effect. First, the radius of the inner rings must exhibit a power-law decay of the type

$$r(x) = (-1)^n \frac{R}{L^n} x^n, \quad (2)$$

with n being a positive integer (note that $r(-L) = R$ and $r(0) = 0$). Second, the wall admittance of the structure $Y(x)$ must have an appropriate dependence with $r(x)$. If one neglects the influence of the inner ring thicknesses and simply considers the air cavities, the wall admittance can be well approximated by the continuous lumped admittance (see Section 3.1):

$$Y(x) = -j \frac{k_0 R^2 - r^2}{Z_0 2r}. \quad (3)$$

Substituting (2) and (3) into (1), and taking into account that $S = \pi r^2$, results in

$$\frac{\partial^2 p}{\partial x^2} + \frac{2n}{x} \frac{\partial p}{\partial x} + \left(\frac{k_0 L^n}{x^n} \right)^2 p = 0. \quad (4)$$

In this work we will be interested in the cases $n=1$ (linear ABH) and $n=2$ (quadratic ABH) of the preceding equation.

2.2. The linear acoustic black hole

2.2.1. Analytical solution of the linear ABH

The linear ABH has been analyzed in detail in [12]. In this subsection, we will summarize some of its main results to facilitate comparison with the quadratic ABH and with the developments in Section 3.

Taking $n=1$ in Eq. (4) yields

$$\frac{\partial^2 p}{\partial x^2} + \frac{2}{x} \frac{\partial p}{\partial x} + \frac{k_0^2 L^2}{x^2} p = 0. \quad (5)$$

The general solution to (5) is given by

$$p(x) = C_+ \exp[\alpha_+ \ln x] + C_- \exp[\alpha_- \ln x] \quad (6)$$

with C_+ , C_- standing for real constants and

$$\alpha_{\pm} = \frac{1}{2} \left[-1 \pm \sqrt{1 - (2k_0 L)^2} \right]. \quad (7)$$

Note that (6) is nothing but the power-law solution in [12], though we have written it in this form for the convenience of forthcoming developments. From (7) it becomes clear that whenever $k_0 L > 0.5$, $\alpha_{\pm} = (1/2) \{ -1 \pm j[(2k_0 L)^2 - 1]^{1/2} \}$ and the solution (6) will exhibit an oscillatory behavior. If we focus, for instance, on the first term in (6), the imaginary part of its

exponent can be understood as that corresponding to a wave propagating to the right with growing amplitude as $x \rightarrow 0$, and with local wavenumber $k(x)$ such that

$$\int_{-L}^x k(x) dx = \frac{1}{2} \ln x \sqrt{(2k_0 L)^2 - 1}. \quad (8)$$

Therefore we get

$$k(x) = \frac{1}{2x} \sqrt{(2k_0 L)^2 - 1}. \quad (9)$$

The group velocity for a wave packet entering the retarding structure at $x = -L$ can be computed as

$$c_g = \left(\frac{\partial k}{\partial \omega} \right)^{-1} = \frac{c_0^2 x}{2L^2 \omega} \sqrt{(2k_0 L)^2 - 1}, \quad (10)$$

and the time it would take for the wavepacket to reach the end of the duct would be

$$T = \lim_{l \rightarrow 0^-} \int_{-L}^{-l} \frac{dx}{c_g} = \lim_{l \rightarrow 0^+} \frac{L^2 \omega}{2c_0^2 \sqrt{(2k_0 L)^2 - 1}} \ln \left(\frac{L}{l} \right) \rightarrow \infty. \quad (11)$$

In other words, a wave entering the structure would never reach its end so no reflection can occur at $x=0$.

2.2.2. Reflection coefficient of the linear ABH

It is clear from (3) that the wall admittance $Y(x)$ becomes singular at $r(0) = 0$. Similarly, Eqs. (6) and (7) reveal that the acoustic pressure blows up at the origin. Consequently, one has to assume a small imperfection in the ABH to describe it with the above presented formalism. The retarding structure must end at a finite distance $x = -l$ from the origin, with $-L < x < -l < 0$, but never reach it. l can be taken as small as wanted. For a perfect linear ABH, the reflection coefficient at $x = -L$, $\mathcal{R}_L^{\text{lin}}$, would be zero, yet this will not be the case for the ABH with the finite distance imperfection. As shown in [12], $\mathcal{R}_L^{\text{lin}}$ can be computed imposing acoustic pressure and particle velocity continuity at $x = -L$, as well as a boundary condition at section $x = -l$, which we presume to have admittance Y_l . The reflection coefficient $\mathcal{R}_L^{\text{lin}}$ is then given by

$$\mathcal{R}_L^{\text{lin}} = \frac{1 + \mathcal{R}_l^{\text{lin}} + \frac{1}{jk_0 L} (\alpha_+ + \mathcal{R}_l^{\text{lin}} \alpha_-)}{1 + \mathcal{R}_l^{\text{lin}} - \frac{1}{jk_0 L} (\alpha_+ - \mathcal{R}_l^{\text{lin}} \alpha_-)} e^{-j2k_0 L}, \quad (12)$$

with $\mathcal{R}_l^{\text{lin}}$ standing for the reflection coefficient at $x = -l$,

$$\mathcal{R}_l^{\text{lin}} = - \frac{\alpha_+ + jk_0 Z_0 l Y_l}{\alpha_- + jk_0 Z_0 l Y_l} \left(\frac{l}{L} \right)^{\alpha_+ - \alpha_-}. \quad (13)$$

In the forthcoming subsection we will extend the preceding results to the quadratic ABH.

2.3. The quadratic acoustic black hole

2.3.1. Analytical solution of the quadratic ABH

The equation governing the behavior of the quadratic ABH results from taking $n=2$ in (4):

$$\frac{\partial^2 p}{\partial x^2} + \frac{4}{x} \frac{\partial p}{\partial x} + \frac{k_0^2 L^4}{x^4} p = 0, \quad (14)$$

and has a general solution of the type

$$p(x) = C_+ \left(-1 + j \frac{k_0 L^2}{x} \right) \exp \left(j \frac{k_0 L^2}{x} \right) + C_- \left(1 + j \frac{k_0 L^2}{x} \right) \exp \left(-j \frac{k_0 L^2}{x} \right), \quad (15)$$

C_+ and C_- denoting real constants. Expression (15) can be rewritten as

$$p(x) = \frac{C_+}{x} \sqrt{(k_0 L^2)^2 + x^2} \exp \left\{ j \left[\frac{k_0 L^2}{x} - \arctan \left(\frac{k_0 L^2}{x} \right) \right] \right\} + \frac{C_-}{x} \sqrt{(k_0 L^2)^2 + x^2} \exp \left\{ -j \left[\frac{k_0 L^2}{x} - \arctan \left(\frac{k_0 L^2}{x} \right) \right] \right\}, \quad (16)$$

the first term representing a wave of growing amplitude as it approaches the duct termination at $x=0$, and having a local

wavenumber that fulfills

$$\int_{-L}^x k(x) dx = \frac{k_0 L^2}{x} - \arctan\left(\frac{k_0 L^2}{x}\right), \quad (17)$$

so that

$$k(x) = \frac{1}{x^2} \frac{(k_0 L^2)^3}{(k_0 L^2)^2 + x^2}. \quad (18)$$

From (18) we can obtain the group velocity for a wave packet entering the quadratic ABH,

$$c_g = \left(\frac{\partial k}{\partial \omega}\right)^{-1} = \frac{1}{L^6 \omega^2} \frac{c_0 x^2 (x^2 c_0^2 + L^4 \omega^2)^2}{3x^2 c_0^2 + L^4 \omega^2}, \quad (19)$$

so that the time it will take for the wavepacket to reach the duct termination at $x=0$ will be again infinite, as for the linear ABH,

$$T = \lim_{l \rightarrow 0^-} \int_{-L}^{-l} \frac{dx}{c_g} = \lim_{l \rightarrow 0^+} \left[\frac{L^6 \omega^2}{c_0 l (L^4 \omega^2 + c_0^2 l^2)} - \frac{L^6 \omega^2}{c_0 L (L^4 \omega^2 + c_0^2 L^2)} \right] \rightarrow \infty. \quad (20)$$

2.3.2. Reflection coefficient of the quadratic ABH

To obtain the reflection coefficient $\mathcal{R}_L^{\text{quad}}$ of the quadratic ABH we shall proceed analogously to what was done in [12] for the linear one. Imposing pressure continuity at $x = -L$ of plane waves propagating in the duct with waves in the retarding structure provides

$$e^{-jk_0 L} + \mathcal{R}_L^{\text{quad}} e^{jk_0 L} = C_+ (-1 - jk_0 L) e^{-jk_0 L} + C_- (1 - jk_0 L) e^{jk_0 L}. \quad (21)$$

On the other hand, momentum conservation for harmonic motion implies $\partial p / \partial x = \rho_0 j \omega v$, so that continuity of the velocity v at $x = -L$ results in

$$e^{-jk_0 L} - \mathcal{R}_L^{\text{quad}} e^{jk_0 L} = (C_+ e^{-jk_0 L} - C_- e^{jk_0 L}) j k_0 L. \quad (22)$$

Once more, we will have to assume an imperfect ABH ending at $x = -l$ because the solution (15) is singular at $x=0$. If Y_l represents the admittance at section $x = -l$, the incident plus outgoing pressure must equal v/Y_l yielding

$$\begin{aligned} C_+ \left(-1 - j \frac{k_0 L^2}{l} \right) \exp\left(-j \frac{k_0 L^2}{l}\right) + C_- \left(1 - j \frac{k_0 L^2}{l} \right) \exp\left(j \frac{k_0 L^2}{l}\right) \\ = - \frac{k_0 L^4}{j Z_0 l^3 Y_l} \left[C_+ \exp\left(-j \frac{k_0 L^2}{l}\right) - C_- \exp\left(j \frac{k_0 L^2}{l}\right) \right]. \end{aligned} \quad (23)$$

Therefore, the reflection coefficient at $x = -l$ can be directly obtained from (23),

$$\mathcal{R}_l^{\text{quad}} = \frac{C_-}{C_+} = \frac{1 + j \frac{k_0 L^2}{l} \left(1 + \frac{L^2}{Z_0 l^2 Y_l} \right)}{1 - j \frac{k_0 L^2}{l} \left(1 - \frac{L^2}{Z_0 l^2 Y_l} \right)} e^{-2j \frac{k_0 L^2}{l}}. \quad (24)$$

The reflection coefficient at $x = -L$ can be derived from the quotient of (21)–(22) by (21) + (22) and considering (24). We get,

$$\mathcal{R}_L^{\text{quad}} = \frac{\mathcal{R}_l^{\text{quad}} e^{jk_0 L} - e^{-jk_0 L} (1 + 2jk_0 L)}{\mathcal{R}_l^{\text{quad}} e^{jk_0 L} (1 - 2jk_0 L) - e^{-jk_0 L}} e^{-2jk_0 L}. \quad (25)$$

3. The transfer matrix approach

3.1. Transfer matrices for a cavity plus ring ensemble

As exposed in Sections 2.2.2 and 2.3.2, assuming that the retarding structure ends at $x = -l$ constitutes a first imperfection of the ABH. However, when trying to build an ABH in practice much more limitations are encountered. As stated in the Introduction, those are related to the number and size of the rings and cavities used to build the structure, to their thickness, etc. As a consequence, the wall admittance of the waveguide can deviate from the theoretical value in (3),

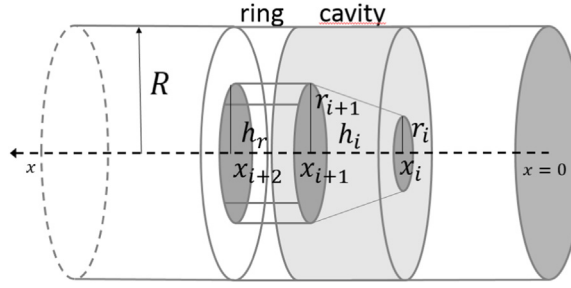


Fig. 2. Sketch of a cavity plus ring ensemble.

resulting in reflection coefficients different from that in (12) for the linear ABH and that in (25) for the quadratic one.

The TMM for acoustic filters can provide a closer model to describe the behavior of the realistic linear and quadratic retarding structures than the analytical ones. To that purpose, one option is to represent the structures as a set of lateral cavities connected by short cylindrical ducts of decreasing radius. The ducts comprise the thicknesses of the ring inner radii, whereas the cavities amount to the space between two consecutive rings (see Fig. 2).

Actually, the TMM allows one to relate the state vector for the acoustic pressure and acoustic particle velocity at a location x_m in the waveguide, namely $(p_m, v_m)^T$, with the state vector at any other location in the waveguide. Easier expressions for the transfer matrices can be found, however, if one works with the acoustic volume velocity u_m , instead of with the particle velocity v_m . Therefore, the state vector, $(p_m, u_m)^T$, and matrices relating its value at different positions will be used in the forthcoming expressions. At a given location one can always switch between u_m and v_m taking into account that $u_m = S_m v_m$, S_m being the section of the waveguide at x_m .

The state vector $(p_{i+2}, u_{i+2})^T$ at x_{i+2} (end of the inner ring edge, see Fig. 2) can be linked to $(p_{i+1}, u_{i+1})^T$ at x_{i+1} (beginning of the inner ring edge) by means of a propagation transfer matrix $\mathbf{T}_{i+1}^{\text{ring}}$. Similarly, the acoustic pressure and acoustic volume velocity $(p_{i+1}, u_{i+1})^T$ at x_{i+1} , which also corresponds to the end of the lateral cavity in Fig. 2, can be related to those at its beginning, $(p_i, u_i)^T$, by means of another transfer matrix $\mathbf{T}_i^{\text{cav}}$. $\mathbf{T}_i^{\text{cav}}$ has to account for three effects. First, the influence of the cavity itself, which can be represented by means of a lumped element with transfer matrix, say $\mathbf{T}_i^{\text{lm}}$. Second, the change in section between the entrance and exit of the cavity. When using the volume velocity as an acoustic variable this effect is automatically considered. Third and last, its finite width, which can be included by means of a propagating matrix, say $\mathbf{T}_i^{\text{p}}$, analogous to that used for the ring. Altogether leads to

$$\begin{aligned}
 \begin{pmatrix} p_{i+2} \\ u_{i+2} \end{pmatrix} &= \mathbf{T}_{i+1}^{\text{ring}} \begin{pmatrix} p_{i+1} \\ u_{i+1} \end{pmatrix} = \mathbf{T}_{i+1}^{\text{ring}} \mathbf{T}_i^{\text{cav}} \begin{pmatrix} p_i \\ u_i \end{pmatrix} = \mathbf{T}_{i+1}^{\text{ring}} \mathbf{T}_i^{\text{p}} \mathbf{T}_i^{\text{lm}} \begin{pmatrix} p_i \\ u_i \end{pmatrix} \\
 &= \begin{pmatrix} \cos(k_0 h_r) & j \frac{Z_0}{S_{i+1}} \sin(k_0 h_r) \\ j \frac{S_{i+1}}{Z_0} \sin(k_0 h_r) & \cos(k_0 h_r) \end{pmatrix} \begin{pmatrix} \cos(k_0 h_i) & j \frac{Z_0}{S_{i+1}} \sin(k_0 h_i) \\ j \frac{S_{i+1}}{Z_0} \sin(k_0 h_i) & \cos(k_0 h_i) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y_i^{\text{cav}} & 1 \end{pmatrix} \begin{pmatrix} p_i \\ u_i \end{pmatrix} \\
 &= \begin{pmatrix} \cos[k_0(h_r + h_i)] + j \frac{Z_0 Y_i^{\text{cav}}}{S_{i+1}} \sin[k_0(h_r + h_i)] & j \frac{Z_0}{S_{i+1}} \sin[k_0(h_r + h_i)] \\ j \frac{S_{i+1}}{Z_0} \sin[k_0(h_r + h_i)] + Y_i^{\text{cav}} \cos[k_0(h_r + h_i)] & Y_i^{\text{cav}} \cos[k_0(h_r + h_i)] \end{pmatrix} \begin{pmatrix} p_i \\ u_i \end{pmatrix} \\
 &\equiv \mathbf{T}_{i+1}^{\text{rc}} \begin{pmatrix} p_i \\ u_i \end{pmatrix},
 \end{aligned} \tag{26}$$

where explicit expressions for $\mathbf{T}_{i+1}^{\text{ring}}$, $\mathbf{T}_i^{\text{p}}$ and $\mathbf{T}_i^{\text{lm}}$ are presented in the second line, and in the last one we have identified the matrix $\mathbf{T}_{i+1}^{\text{rc}}$ that characterizes the effect of the ensemble cavity plus ring on the propagating acoustic pressure and acoustic volume velocity.

In (26), $h_i = |x_{i+1} - x_i|$ stands for the cavity width and $h_r = |x_{i+2} - x_{i+1}|$ for the ring thickness (assumed constant for all rings in this work). In the limiting case of negligible thickness, i.e., $h_r \approx 0$, $\mathbf{T}_{i+1}^{\text{ring}}$ simply becomes the identity matrix and the ring shows no influence on the retarding structure behavior. This is what is in fact assumed in the governing equation (4) of the ABH. Besides, in matrices $\mathbf{T}_i^{\text{lm}}$ and $\mathbf{T}_{i+1}^{\text{ring}}$, $S_{i+1} = \pi r_{i+1}^2$ with r_{i+1} standing for the inner radius of the ring at x_{i+1} , and the admittance Y_i of the cavity is approximated by

$$Y_i^{\text{cav}} = j \frac{k_0}{Z_0} V_i. \tag{27}$$

The volume V_i has been taken as that of a truncated cone (see Fig. 2) which extends from coordinate x_i to x_{i+1} :

$$V_i = \pi h_i \left[R^2 - \frac{1}{3} (r_{i+1}^2 + r_i^2 + r_{i+1} r_i) \right]. \quad (28)$$

This volume has inner virtual surface

$$S_i^{\text{con}} = \pi (r_{i+1} + r_i) \sqrt{h_i^2 + (r_{i+1} - r_i)^2}, \quad (29)$$

so that taking the limit we get

$$\lim_{x_{i+1} \rightarrow x_i} \frac{Y_i^{\text{cav}}}{S_i^{\text{con}}} = -j \frac{k_0}{Z_0} \lim_{x_{i+1} \rightarrow x_i} \frac{\pi h_i \left[R^2 - \frac{1}{3} (r_{i+1}^2 + r_i^2 + r_{i+1} r_i) \right]}{\pi (r_{i+1} + r_i) \sqrt{h_i^2 + (r_{i+1} - r_i)^2}} = -j \frac{k_0}{Z_0} \frac{R^2 - r_i^2}{2r_i}, \quad (30)$$

which is nothing but the value of the local admittance (3) used in the governing equation of the analytical ABH. Note also that in (27) it is assumed that $k_0 r_i \ll 1$. Given the size of the rings and cavities, one could have also made the approximations $\cos(k_0 h_r) \sim \cos(k_0 h_i) \sim 1$ and $\sin(k_0 h_r) \sim k_0 h_r$, $\sin(k_0 h_i) \sim k_0 h_i$, in (26).

The easiest way to introduce damping into the system is by considering a constant complex sound of speed $c = c_0(1 + \mu j)$ (with μ a real positive constant) as in [12], or analogously, by using a complex wave number $k = k_0(1 - \mu j)$ instead of k_0 . However, in practical realizations of the ABH one would typically resort to filling the cavities with some sound absorbing material. Given that the cavities are separated by rings, waves propagating through the lining in the waveguide axis direction can be discarded and we can assume the material to be locally reacting. Propagation within the lining can be described by means of a characteristic impedance $\tilde{Z}$ and a wavenumber $\tilde{k}$. For a fibrous material, $\tilde{Z}$ and $\tilde{k}$ can be well approximated by the empirical expressions of [35], improved by [36] (also available in [37]):

$$\frac{\tilde{Z}}{Z_0} = \begin{cases} 1 + 0.0485E^{-0.754} - j0.087E^{-0.73} & E < 1/60 \\ [0.5/(\pi E) + j1.4]/(-1.466 + j0.212/E)^{1/2} & E > 1/60, \end{cases} \quad (31)$$

and

$$\frac{\tilde{k}}{k_0} = \begin{cases} 1 - j0.189E^{-0.6185} + 0.0978E^{-0.6929} & E < 1/60, \\ (1.466 - j0.212/E)^{1/2} & E > 1/60, \end{cases} \quad (32)$$

where the absorber variable $E \equiv Z_0 k_0 / (2\pi\sigma)$, σ standing for the air flow resistivity. $\tilde{Z}$ and $\tilde{k}$ should respectively replace Z_0 and k_0 in (27), so that $\mathbf{T}_i^{\text{lm}}$ gets modified when considering a cavity filled with absorbent. As a consequence, the plane acoustic waves traveling within the waveguide will experience an approximate wall impedance at a cavity boundary point x_i (see e.g., [31]):

$$Z_i^w \sim -j\tilde{Z}\cot[\tilde{k}(R - r_i)]. \quad (33)$$

The wavenumber k_0 in $\mathbf{T}_i^{\text{p}}$ of (26) should then be replaced by the complex expression

$$k_i^x \simeq k_0 \sqrt{1 - j \frac{Z_0}{Z_i^w} \frac{2}{k_0 r_i}}, \quad (34)$$

and the admittance Z_0 with

$$Z_i^x \simeq Z_0 \frac{k_0}{k_i^w}, \quad (35)$$

to account also for the effects of the absorbent material in the i -th cavity.

3.2. Transfer matrix of the retarding structure

Equipped with the results in the previous subsection, it becomes straightforward to relate the state vector at an arbitrary section x_{k+2} linked through cavity plus ring ensembles with that at x_i , ($k \geq i$), by means of successive products of matrices $\mathbf{T}_{i+1}^{\text{rc}}$. Let us introduce the transfer matrix

$$\mathbf{A}(k+2, i) \equiv \prod_{m=i}^k \mathbf{T}_{m+1}^{\text{rc}} \quad (36)$$

so that

$$\begin{pmatrix} p_{k+2} \\ u_{k+2} \end{pmatrix} = \mathbf{A}(k+2, i) \begin{pmatrix} p_i \\ u_i \end{pmatrix}. \quad (37)$$

In particular, this allows one to characterize the entire retarding structure. For a total of N rings, the state vector at the entrance $x_N = -L$ of the retarding structure can be connected to that at its termination $x_0 = 0$ by

$$\begin{pmatrix} p_N \\ u_N \end{pmatrix} = \mathbf{A}(N, 0) \begin{pmatrix} p_0 \\ u_0 \end{pmatrix}. \quad (38)$$

3.3. Reflection coefficient of the retarding structure

The duct admittance at the entrance of the retarding structure $x = -L$ can be computed according to (38) as

$$Y_L = \frac{u_N}{p_N} = \frac{A_{21}p_0 + A_{22}u_0}{A_{11}p_0 + A_{12}u_0} = \frac{A_{21} + A_{22}Y_{l=0}}{A_{11} + A_{12}Y_{l=0}}, \quad (39)$$

$Y_{l=0}$ standing for the admittance at $x=0$. The reflection coefficient at $x = -L$ is related to Y_L through

$$\mathcal{R}_L^{\text{TM}} = \frac{\pi R^2 - Z_0 Y_L}{\pi R^2 + Z_0 Y_L}. \quad (40)$$

The reflection coefficient $\mathcal{R}_L^{\text{TM}}$ obtained from the transfer matrix method is expected to better characterize the performance of the realistic retarding structure than its continuous counterparts because it includes several aspects the latter cannot taken into account. $\mathcal{R}_L^{\text{TM}}$ is to be compared with $\mathcal{R}_L^{\text{lin}}$ in (12) and $\mathcal{R}_L^{\text{quad}}$ in (25) for the linear and quadratic ABHs.

4. Numerical simulations

4.1. Influence of imperfections on the analytical linear and quadratic ABHs

Let us consider a cylindrical duct of radius $R=0.23$ m, with a retarding structure ($L=0.5$ m in length) placed at its termination. The cutoff frequency of the duct is given by $f_c = 1.84c_0/R = 2720$ Hz, with $c_0 = 340$ m/s. Losses are introduced into the system by means of a complex wavenumber $c = c_0(1 + 0.05j)$.

To start with, we contemplate the hypothetical situation in which there is no reflection at all from the last section, $x = -L$, of the retarding structure. In other words, we directly assume $\mathcal{R}_L^{\text{lin}} = 0$ in (13) and $\mathcal{R}_L^{\text{quad}} = 0$ in (24), for the linear and quadratic ABHs respectively. The reflection coefficients $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ at the entrance of the retarding structure, $x = -L$, will be given by (12) in the linear case and by (25) in the quadratic one. The dependence of $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ with the non-dimensional number k_0L has been plotted in Fig. 3a. Note that according to the duct cutoff frequency results will be only valid up to $k_0L \sim 4$.

As observed, and as expected from (7), for $k_0L < 0.5$ the linear ABH is not operative and reflects almost all incident sound, so that $\mathcal{R}_L^{\text{lin}} \sim 1$. For $0.6 < k_0L < 3$ the linear ABH performs slightly better than the quadratic one. However, the situation reverses as long as one takes into account reflections from $x = -l$. Suppose now that the duct termination is totally rigid. The acoustic particle velocity will vanish at it and $Y_l=0$. Whilst the imperfection length l remains very tiny this will not make a big difference. The reflection surface at the end of the duct will be $\pi R^2(l/L)^2$ in the linear case and $\pi R^2(l/L)^4$ for the quadratic one. For a very small l these surfaces will be negligible and as one can readily check from (13) and (24), it turns out that $\mathcal{R}_L^{\text{lin}} \approx \mathcal{R}_L^{\text{quad}} \approx 0$. Yet, if we increase a little bit the imperfection up to $l = 5 \times 10^{-5}$ m, which is still a very small value for manufacturing purposes, the behavior of the linear ABH clearly deteriorates. Peaks and deeps appear for $\mathcal{R}_L^{\text{lin}}$, whereas $\mathcal{R}_L^{\text{quad}}$ remains almost unaltered (see Fig. 3b). When we enlarge the value of the imperfection up to $l = 5 \times 10^{-3}$ m the situation becomes more apparent and the quadratic ABH becomes clearly the best option. This can be observed in Fig. 3c. The imperfection needs to be as high as $l = 5 \times 10^{-2}$ m for $\mathcal{R}_L^{\text{quad}}$ to begin to oscillate in Fig. 3d. In other words, the quadratic ABH is clearly more robust to imperfections. A reasonable explanation for this behavior is as follows. The radius of the quadratic ABH approaches the duct termination at $x=0$ with a parabolic profile, which is smoother than the straight line of the linear ABH. Therefore, suppressing a small portion of the linear ABH has a bigger impact on the latter than suppressing the same portion on the quadratic ABH.

4.2. Influence of the number of rings

As commented before, when aiming at a practical realization of the ABH in Fig. 1 (see e.g., [13,14]) several aspects are to be taken into account. First, one could wonder, for example, how many rings and of which thickness are necessary to try to recover the results predicted by the analytical formulation. The transfer matrix approach in Section 3 offers a rapid way to address those issues. Let us start with the influence the number of rings has on the reflection coefficient of the retarding structure in the preceding subsection.

To separate the effects of the various parameters playing a role in the ABH performance, we consider herein the case of very thin rings of thickness $h_r = 1$ μm , so that $\mathbf{T}_{i+1}^{\text{ring}}$ in (26) nearly becomes the identity matrix. Equal spaced rings of constant thickness will be assumed throughout this work. The influence of the number of rings is shown in Figs. 4a and b for the

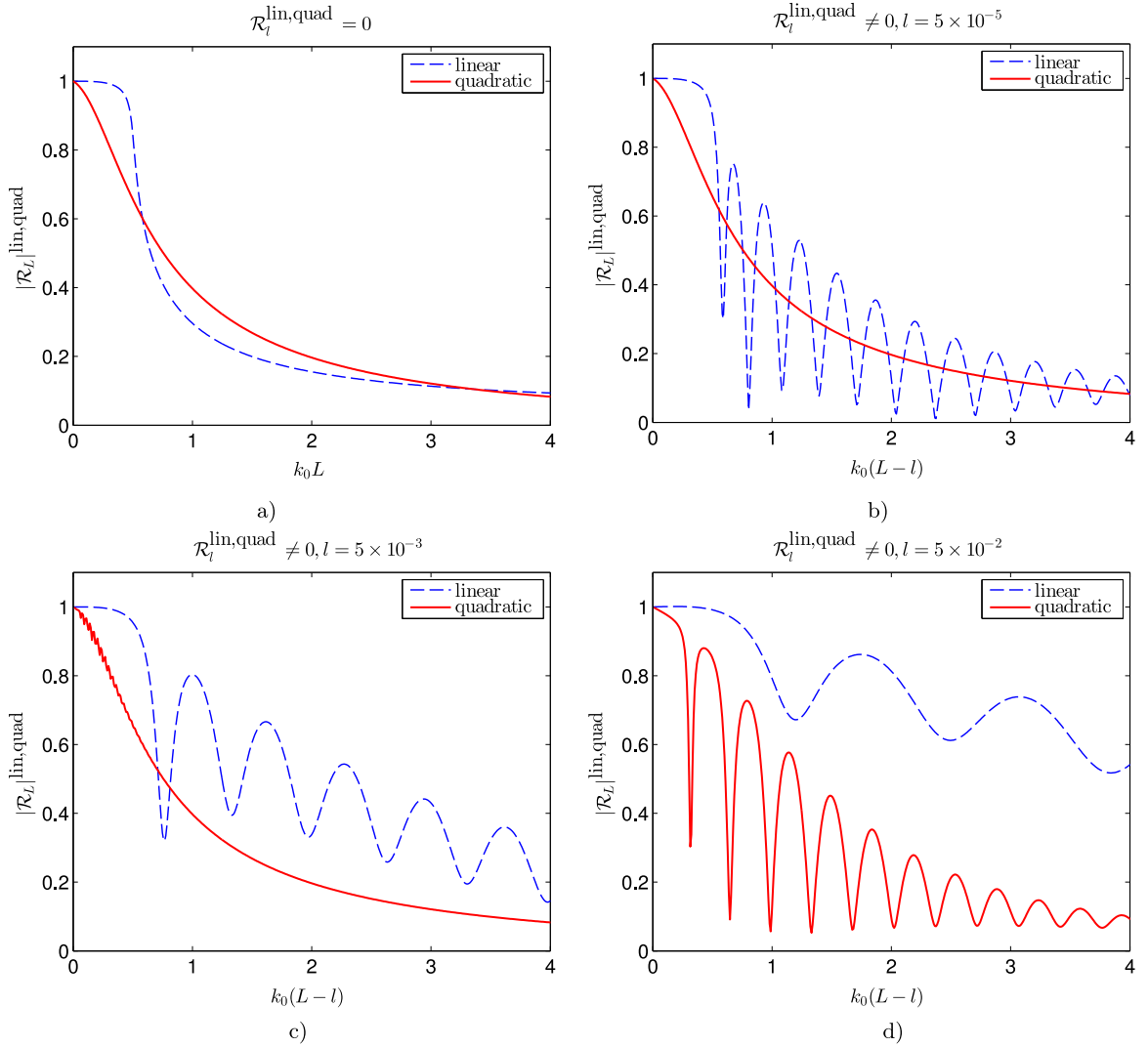


Fig. 3. Analytical reflection coefficients $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ for the linear and quadratic ABHs. (a) No reflection from the duct termination, (b) rigid termination with $l = 5 \times 10^{-5}$ m, (c) rigid termination with $l = 5 \times 10^{-3}$ m, and (d) rigid termination with $l = 5 \times 10^{-2}$ m.

linear ABH, where the reflection coefficient $\mathcal{R}_L^{\text{TM}}$ is plotted respectively for imperfect ABHs with $l = 10^{-3}$ m and $l = 3 \times 10^{-3}$ m. Analogous results are presented in Figs. 4c and d for the quadratic ABH. The figures also include the analytical reflection coefficients $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ for comparison (black continuous line in the figures). As expected, $\mathcal{R}_L^{\text{TM,lin}}$ and $\mathcal{R}_L^{\text{TM,quad}}$ tend to the analytical curves $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ when increasing the number N of rings, but they do so at different ratios. For small N (20 and 40) the approximations are very coarse for both, the linear and quadratic cases. However, when we increase N an order of magnitude the reflection coefficient of the quadratic ABH quickly approximates the analytic one. The convergence is much weaker in the linear case. The high number of rings needed to recover the analytic ABHs may pose a severe limitation to practical realizations of the ABH, which aim at a limited number of rings for manufacturing purposes.

4.3. Influence of the ring thickness

In Figs. 5a and b we show the influence of the ring thickness on the reflection coefficient $\mathcal{R}_L^{\text{TM}}$ of the retarding structure, for the linear and quadratic cases. To proceed, the number of rings has been fixed to $N=40$ and we have varied h_r from the very small value of $h_r = 1 \mu\text{m}$ to those of $h_r = 1, 2, 4$ mm (2 mm thickness rings were used, for instance, in the retarding structures built in [13,14]). Figs. 5a and b respectively present the results for a linear and a quadratic ABH for an imperfection of $l = 10^{-3}$ m. Results for $l = 3 \times 10^{-3}$ m looked very similar to those of $l = 10^{-3}$ m, though performing slightly worse as expected, and therefore will not reproduce it herein. As noticed, the ring thickness does not substantially affect the amplitude of the peaks and dips in $\mathcal{R}_L^{\text{TM}}$ but their frequency location. The deviations become larger as $k_0(L-l)$ increases. Unless

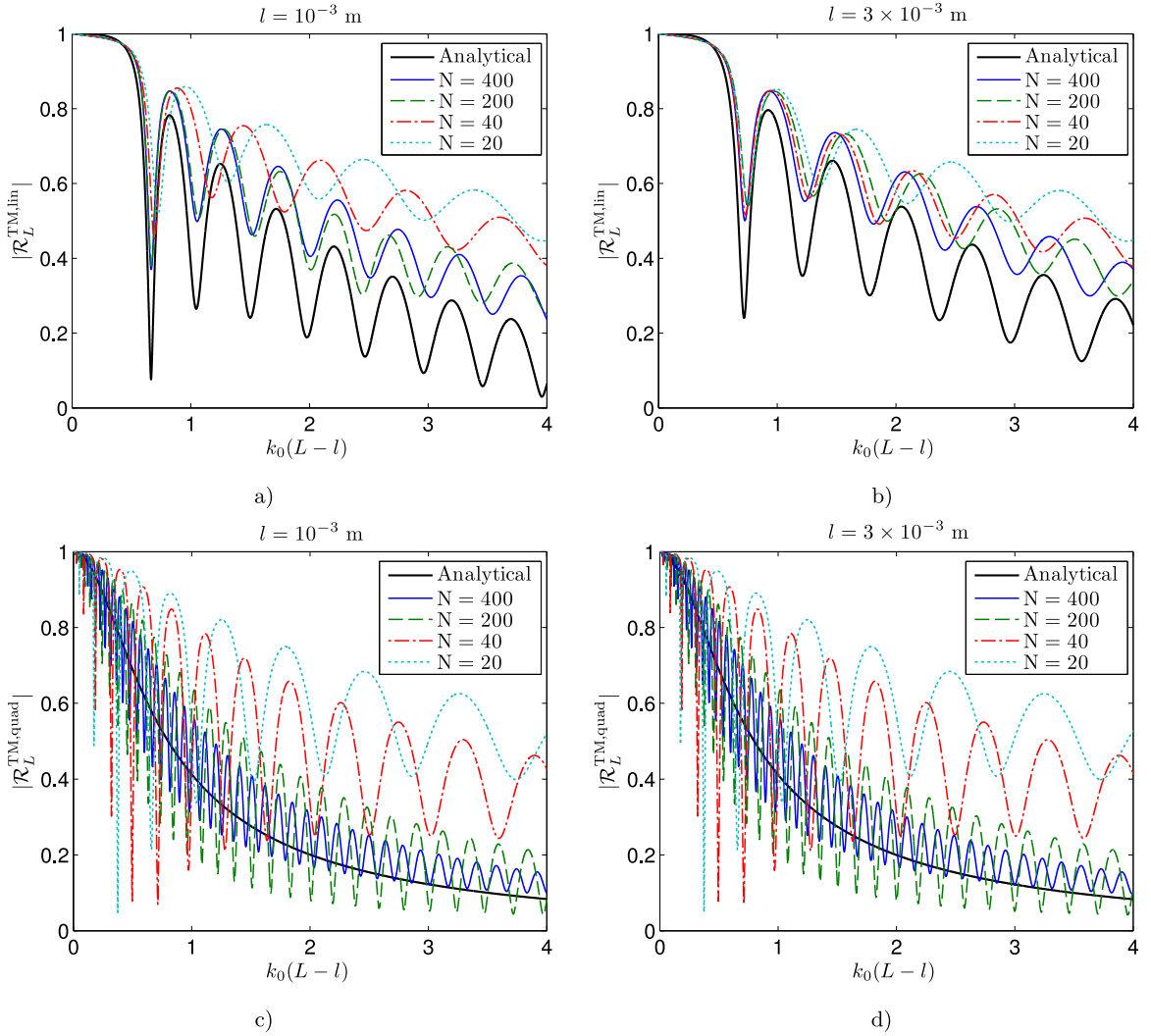


Fig. 4. Influence of the number of rings on $\mathcal{R}_L^{\text{TM}}$ for linear and quadratic ABHs. (a) Linear ABH with $l = 10^{-3}$ m, (b) linear ABH with $l = 3 \times 10^{-3}$ m, (c) quadratic ABH with $l = 10^{-3}$ m, and (d) quadratic ABH with $l = 3 \times 10^{-3}$ m.

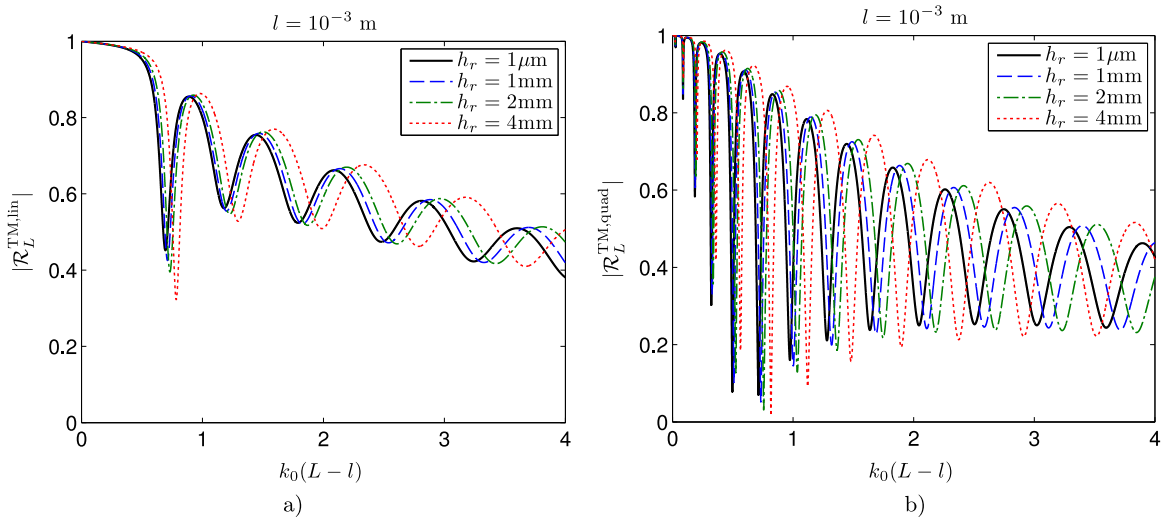


Fig. 5. Influence of the ring thickness on $\mathcal{R}_L^{\text{TM}}$ for linear (a) and quadratic (b) ABHs, $N=40$ rings and $l = 10^{-3}$ m.

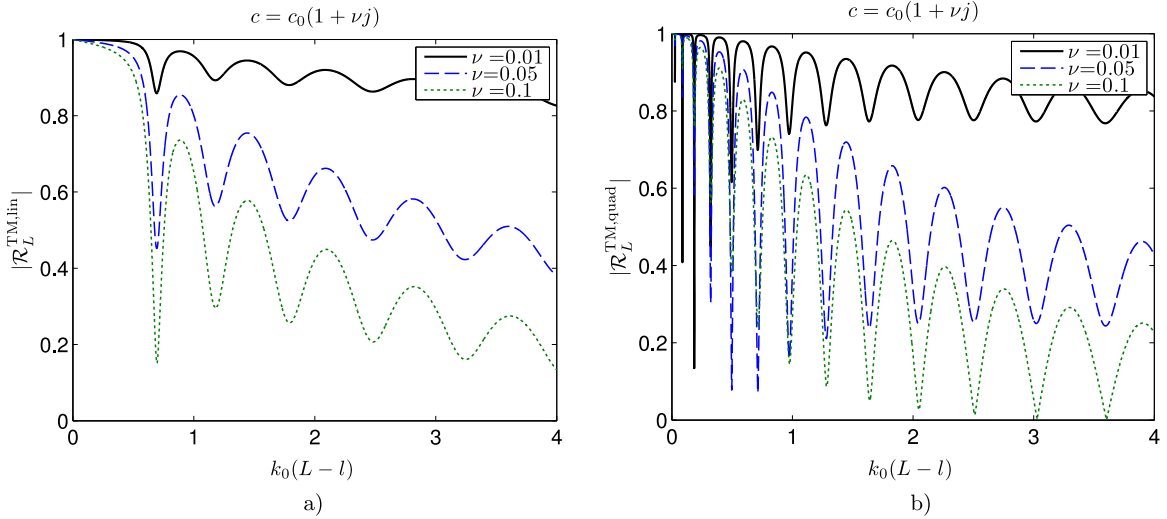


Fig. 6. Influence of damping with complex sound speed on $\mathcal{R}_L^{\text{TM}}$ for linear (a) and quadratic (b) ABHs, $N=40$ rings and $l = 10^{-3}$ m.

one has the problem of requiring special attenuation at a given particular frequency, Fig. 5 shows that the $\mathcal{R}_L^{\text{TM}}$ is quite robust to variations in the ring thickness.

4.4. Influence of damping

The effects of damping on $\mathcal{R}_L^{\text{TM}}$ have been first analyzed in Figs. 6a and b for the linear and quadratic ABHs, when considering a complex speed of sound $c = c_0(1 + \mu j)$, with values $\mu = \{0.01, 0.05, 0.1\}$. The number of rings has been fixed again to $N=40$, the ring thickness has been set to $h_r = 1 \mu\text{m}$ and we have considered an imperfection of $l = 10^{-3}$ m. Figs. 6a and b reveal damping plays a crucial role in the performance of both, the linear and quadratic ABHs. Though of academic interest, for damping the retarding structure in practice one would rather resort to fill some of the cavities with absorbent material. To that purpose we have considered a fibrous material with air flow resistivity $\sigma = 1000 \text{ Pa s/m}^2$ and make use of the formulation described in Section 3.1.

In Figs. 7a and b we have plotted the absorption coefficient α_n of every cavity in the retarding structure for both, the linear and quadratic ABHs. The absorption coefficient for the cavities respectively located at $x = -0.25L, -0.5L, -0.75L$ and $-L$ have been highlighted to have some reference curves. Obviously α_n is greater for the bigger cavities located close to the ABH termination. The differences in the distribution of the absorption within the waveguide are apparent from the figures of α_n^{lin} and α_n^{quad} . In Figs. 7c and d we present the corresponding reflection coefficients $\mathcal{R}_L^{\text{TM}}$ for the linear and quadratic ABHs. In these figures the percentage is cumulative, i.e., the labels 25% or 50% respectively indicate that all cavities from the termination to $-0.25L$ (resp. $-0.5L$) have been filled with absorbent material. As observed, filling the last 25% portion of the waveguide does not seem to guarantee a low reflection coefficient as one may have suspected a priori, and some uncontrolled oscillations appear for large values of $k_0(L-l)$. The situation is not so strange given that no loss mechanism at all has been considered for the remaining portion of the ABH, which is somewhat unrealistic. The situation improves when increasing the number of cavities replenished with fibrous material. Quite amazingly, the linear ABH performs better than the quadratic one. That behavior would require a deeper analysis in the future. Actually, it is worthwhile mentioning that in [13,14] several configurations with absorbent material were experimentally tested. However, the obtained results were kind of frustrating because the improvement of the ABHs performance was not as expected. The authors of those works claimed an explanation is still missing for that occurrence.

4.5. Influence of ring minimum inner radius

Manufacturing constrictions may not only be due to the ring thickness or ring inter-space, but also to the ring minimum inner hole one could drill. If there is a minimum value for r_i , and we want to keep the power-law decay in (2) unaltered, this implies the retarding structure has to be shortened, see Fig. 8a. Given that the quadratic ABH approaches the duct termination very smoothly, for a fixed value of r_i a large portion of it has to be suppressed in comparison with the linear ABH (i.e., $l_{\text{quad}} \gg l_{\text{lin}}$ in Fig. 8a). We may wonder if the ABH performances could then be improved by directly defining linear and parabolic profiles between points $(0, r_i)$ and $(-L, R)$ (see Fig. 8b), which would slightly vary the decay law in (2). In other words, use is made of

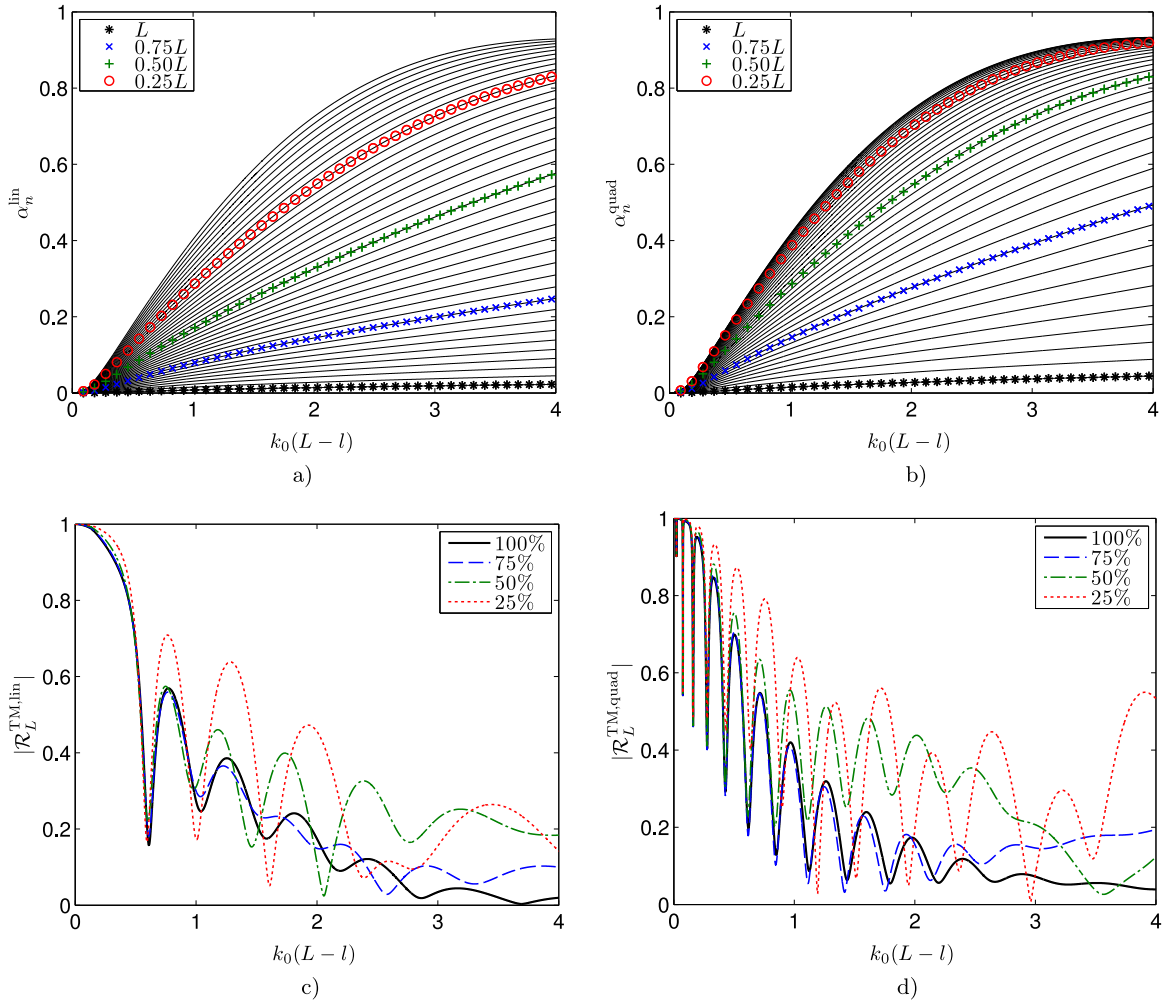


Fig. 7. Influence of absorbent material damping on $\mathcal{R}_L^{\text{TM}}$ for linear and quadratic ABHs. (a) Absorption coefficient for the cavities of the linear ABH, (b) absorption coefficient for the cavities of the quadratic ABH, (c) $\mathcal{R}_L^{\text{TM,lin}}$ for various percentages of lining, and (d) $\mathcal{R}_L^{\text{TM,quad}}$ for various percentages of lining.

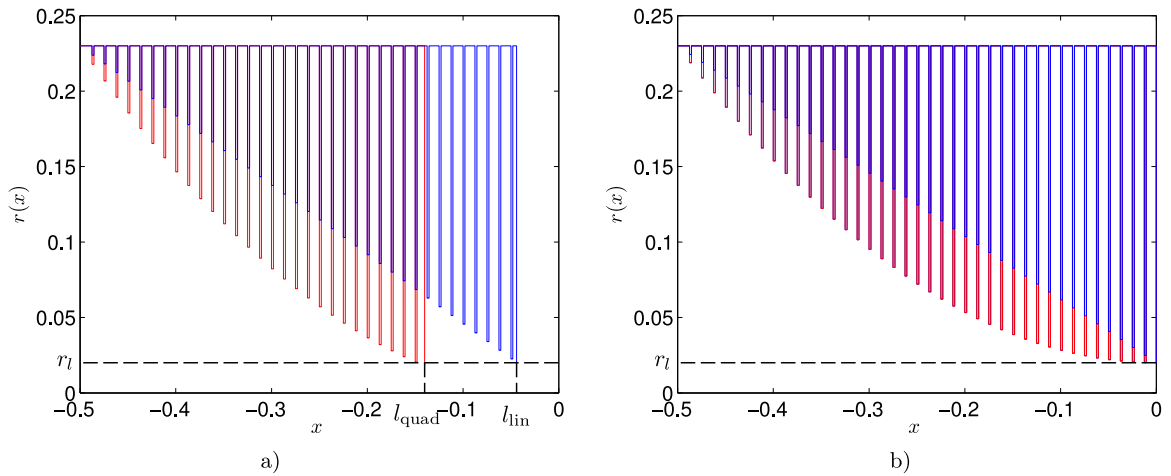


Fig. 8. Linear and quadratic ABH profiles when fixing the minimum ring inner hole. (a) The power-law decay remains unaltered. (b) The power-law decay becomes slightly modified to improve the ABHs performance.

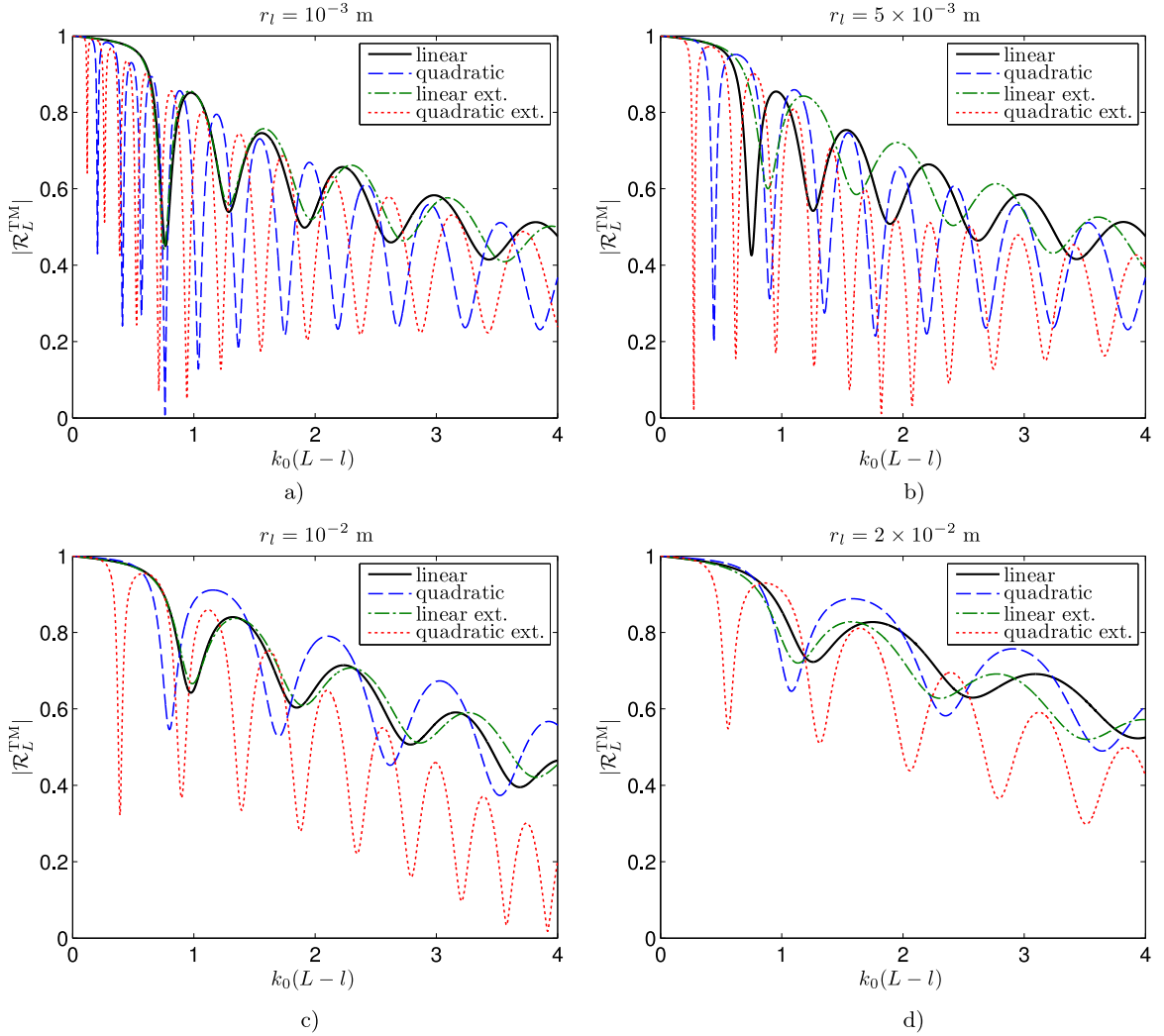


Fig. 9. Reflection coefficients $\mathcal{R}_L^{\text{lin}}$ and $\mathcal{R}_L^{\text{quad}}$ for the ABHs in Figs. 8a and b and different minimum last ring radius r_l : (a) $r_l = 10^{-3}$ m, (b) $r_l = 5 \times 10^{-3}$ m, (c) $r_l = 10^{-2}$ m, and (d) $r_l = 5 \times 10^{-2}$ m.

$$r(x) = (-1)^n \frac{R - r_l}{L^n} x^n + r_l, \quad n = 1, 2, \quad (41)$$

instead of (2).

Figs. 9a–d show the effects of such modifications on the reflection coefficient $\mathcal{R}_L^{\text{TM}}$ for four different values of the minimum radius r_l , namely $r_l = \{10^{-3}, 5 \times 10^{-3}, 10^{-2}, 2 \times 10^{-2}\}$ m. Again we have considered the case of $N=40$ rings with thickness $h_r = 1 \mu\text{m}$. In the legends of Fig. 9 linear and quadratic stand for the ABHs in Fig. 8a, while linear ext. and quadratic ext. for those in Fig. 8b.

For the tested radii, Figs. 9a–d reveal that in the case of linear ABHs there are no substantial differences between the performance of the original ABH and the modified one. As said, the reason is that the number of suppressed rings will not be severe in the linear case. The situation becomes, however, very different for the quadratic ABHs. If r_l is rather small, as in Fig. 9a, the original quadratic ABH performs better than the linear ones and its modified version does not improve $\mathcal{R}_L^{\text{TM}}$ in the mean, but simply shifts the locations of its peaks and valleys. Yet, when increasing r_l the behavior of the quadratic ABH in Fig. 8a begins to quickly deteriorate and better results are provided by the modified one, see Fig. 9b. For large values of r_l , like those in Figs. 9c and d, the benefits of modifying the quadratic ABH become very apparent. The original ABH is so shortened that it even performs worst than the linear ABHs.

4.6. Validity of the results

At this point some considerations should be made regarding the validity of the TMM results. These have been presented for frequencies lower than the cutoff one, f_c , which has restricted the dimensionless number k_0L to ~ 4 in the simulations. Consequently, one could expect a uniform acoustic pressure pattern inside the cavities below this frequency. However, the lumped approximation for lateral cavities in the TMM approach demands the wavelength to be much larger than the cavity size, which becomes almost that of the duct radius for those cavities close to the ABH termination. Therefore, the frequency range of validity of the results in the previous subsections may be somewhat lower to that of $k_0L = 4$. Another possible source of discrepancies between the predicted and real acoustic pressure inside the ABH might be due to the modeling of the initial rings, at the entrance of the ABH. It cannot be plenty justified that the cavities at that zone are well represented by means of lumped side branches, and, probably, expansion chamber elements could be a better option.

An additional aspect that the TMM method cannot take into account is that of the acoustic interaction between cavities at some particular frequencies. This is a well-known phenomenon in ducts with side branches and results in strong dips when computing, for instance, acoustic pressure transfer functions between some points inside the duct. As a curiosity, a typical case in biomechanics is that of the effects produced by the valliculae and piriform fossae in human vocal tract acoustics, see e.g., [38,39].

The above issues, among others like the visco-thermal losses in the thinner cavities of the ABH, could be clarified resorting to detailed FEM simulations. Unfortunately, the later can be very costly if a large number of configurations are to be tested. As exposed in the Introduction section, and despite of its drawbacks, the TMM offers a very fast way to test the effects of parameter variations in the performance of the ABHs. It is then to be viewed as a fast prediction tool rather than a very precise one. Once a few ABH configurations had been selected with the TMM, one could proceed to FEM simulations for a better characterization. However, FEM simulations of the ABH acoustics are deemed out of the scope of this paper and will be presented in future work.

5. Conclusions

This paper suggests studying the acoustic black hole (ABH) effect of retarding structures in duct terminations by means of the transfer matrix method (TMM). This allows one to carry out quick tests to determine the influence of the various parameters that can play a role in practical realizations of the ABHs. In particular, the influence of the number of rings, ring thickness, damping and ring minimum inner radius has been analyzed for both linear and quadratic ABHs. For the latter, analytical results have been derived to complement those of the linear case and to allow comparison with the TMM predictions.

As general tendencies, it has been observed that a large number of rings becomes necessary to recover the reflection coefficients predicted by the theory. This is specially the case for linear ABHs, the quadratic ones performing somewhat better. Besides, the ABH effect has proved quite insensitive to variations in ring thickness for both, linear and quadratic ABHs. Not surprisingly, damping has turned to be crucial to achieve low values of the reflection coefficient. Yet, some disquisitive results have been observed when filling the cavities with absorbent material that would deserve further exploration in the future. To conclude, we have also shown that setting a minimum value for the smallest ring inner radius implies a strong shortening of the quadratic ABH length, if we want to keep the radius power-law decay of the formulation. Better results can be achieved by slightly circumventing the latter and working with the original length of the ABH.

The reported analysis provides first indications on what to consider when building an ABH retarding structure. Forthcoming work will involve testing some of the assumptions involved in this work by means of detailed finite element simulations. To conclude, note that the herein presented TMM approach applied to linear and quadratic ABHs can be straightforwardly extended to test higher-order ABHs with power-law exponents greater than two.

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